

BESANCON THISE, FranceWMO: **072880**Lat: **47.2489N**Lon: **5.9889E**Elev: **1014**StdP: **14.17**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.0	22.9	12.6	10.8	23.1	16.8	13.3	26.1	20.7	48.4	18.2	48.0	3.3	40	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	88.2	68.5	84.8	67.6	81.8	66.5	70.3	83.9	68.9	81.8	67.5	79.2	5.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	99.9	75.3	64.6	94.7	72.8	63.3	90.5	71.7	34.8	84.1	33.5	82.1	32.4	79.6	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.3	11.4	DB	13.8	92.7	5.4	3.5	9.9	95.3	6.7	97.3	3.7	99.3	-0.2	101.9	(4)
(5)			WB	13.1	73.4	5.2	1.8	9.4	74.7	6.4	75.7	3.5	76.8	-0.3	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	36.9	38.9	45.2	51.2	58.2	64.6	68.0	67.3	60.3	53.7	44.0	38.2	(6)	
	DBStd	12.92	7.51	7.35	7.08	6.62	6.25	6.58	5.99	6.13	5.81	6.50	7.18	7.63	(7)	
	HDD50	1580	408	313	178	64	8	0	0	0	1	38	202	368	(8)	
	HDD65	4995	871	731	612	413	225	88	37	46	162	350	631	829	(9)	
	CDD50	2420	2	2	31	102	261	438	558	535	311	155	22	3	(10)	
	CDD65	363	0	0	0	1	14	76	131	117	22	2	0	0	(11)	
	CDH74	3440	0	0	1	24	175	732	1230	1079	184	15	0	0	(12)	
	CDH80	1049	0	0	0	0	26	202	413	381	26	1	0	0	(13)	
(14) Wind	WSAvg	4.1	4.7	4.9	5.1	4.5	4.1	3.9	3.8	3.3	3.4	3.3	3.9	4.7	(14)	
(15) Precipitation	PrecAvg	49.0	3.7	3.6	3.4	3.8	4.4	4.0	4.0	3.9	4.2	4.7	4.8	4.4	(15)	
	PrecMax	62.9	7.6	7.7	12.1	8.7	8.0	8.9	10.5	8.4	7.8	8.5	13.5	7.6	(16)	
	PrecMin	37.4	0.8	0.6	0.7	0.7	1.8	1.5	1.5	0.2	1.1	1.0	1.4	0.2	(17)	
	PrecStd	7.0	2.0	1.9	2.4	2.1	1.7	1.7	1.9	1.9	1.9	2.2	2.6	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	60.9	69.1	77.2	83.4	89.7	93.1	95.7	83.3	76.0	64.8	58.4	(19)	
		MCWB	49.4	49.4	53.6	58.6	64.6	68.6	68.7	69.0	65.8	62.2	55.5	50.9	(20)	
	2%	DB	53.1	56.5	65.2	72.5	78.7	85.2	88.0	88.4	79.2	71.4	60.4	54.2	(21)	
		MCWB	48.5	47.3	51.8	56.7	63.0	68.0	68.7	68.5	65.0	60.8	53.7	49.3	(22)	
	5%	DB	50.8	52.9	61.2	68.3	75.2	81.9	84.7	84.3	75.3	67.6	57.2	51.7	(23)	
		MCWB	47.1	46.0	50.0	54.6	61.0	66.7	67.8	67.5	63.7	59.2	52.4	48.5	(24)	
	10%	DB	47.8	49.7	57.1	64.3	71.4	78.2	81.1	80.4	71.4	64.1	54.3	49.3	(25)	
		MCWB	44.8	44.8	48.3	52.8	59.1	64.9	66.5	66.5	61.8	57.8	50.6	46.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	50.5	55.5	60.7	67.1	71.8	72.1	72.2	68.6	64.3	57.1	52.4	(27)	
		MCDB	53.9	56.8	65.7	74.2	79.2	84.5	85.9	85.3	78.4	72.5	61.1	55.0	(28)	
	2%	WB	49.4	48.6	52.9	57.7	64.5	69.5	70.2	70.2	66.4	62.1	54.8	50.5	(29)	
		MCDB	52.2	54.4	61.5	69.2	75.7	82.0	84.2	84.1	76.2	68.9	59.0	53.1	(30)	
	5%	WB	47.2	47.0	51.2	55.7	62.1	67.7	68.7	68.5	64.6	60.3	52.9	48.7	(31)	
		MCDB	50.2	51.7	58.8	66.3	72.2	79.3	81.8	81.8	73.5	66.1	56.5	51.4	(32)	
	10%	WB	45.1	45.3	49.4	53.7	60.1	65.9	67.2	66.9	62.8	58.4	50.9	46.7	(33)	
		MCDB	47.7	49.3	56.3	62.6	68.7	76.4	78.7	78.2	69.6	63.0	53.8	49.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	12.7	16.1	18.0	18.0	18.5	18.8	18.8	17.3	15.6	11.4	9.7	(35)	
		MCDBR	12.4	19.6	24.2	25.5	24.9	24.4	25.1	26.1	23.5	21.1	16.5	12.4	(36)	
	5% WB	MCWBR	8.5	11.9	12.9	12.4	11.5	10.7	9.6	10.1	11.1	11.5	10.4	8.7	(37)	
		MCWBR	10.9	15.8	20.0	23.0	22.5	22.3	22.6	23.1	20.4	19.0	14.5	11.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.305	0.327	0.356	0.391	0.397	0.403	0.400	0.395	0.373	0.357	0.325	0.303	(40)	
		taud	2.458	2.400	2.348	2.266	2.291	2.309	2.313	2.343	2.385	2.421	2.460	2.460	(41)	
	Ebn at Noon		249	266	273	273	275	273	272	269	265	250	238	235	(42)	
		Edh at Noon	21	27	33	40	40	40	39	36	32	27	21	19	(43)	
(44) All-Sky Solar Radiation	RadAvg		348	610	1003	1390	1618	1898	1818	1577	1212	754	401	282	(44)	
	RadStd		45	97	144	199	166	176	158	142	107	66	44	50	(45)	

Historical Trends

	DBAvg
--	-------